EARLY WARNING FOR HIGH-RISK AIR QUALITY HOURS

A time-aware comparison of classification models for Hong Kong's Air Quality Health Index

HONG KONG | 2021–2024

EPD hourly AQHI + HKO daily weather

73.8% — **Random Forest recall**
at threshold 0.893

61.3% — **Lower alert cost**
vs. no-model baseline

0.683 — **Test PR-AUC**
5-fold CV mean: 0.663

RESEARCH QUESTION

Can recent station readings and AQHI trends warn us one hour before the AQHI reaches 7?

A risk-level alert is more actionable than a point forecast.

DATA AT A GLANCE

617,799 hourly station records

18 monitoring stations

27 engineered features

Target: next-hour AQHI ≥ 7 | Positive class: 1.60%

AQHI history (lags, rolling means, volatility, changes) + time-of-day, station type, and daily temperature, humidity, rainfall and wet-bulb data.

EVALUATION DESIGN

01 Chronological split
Final 20% held out as the test period.

02 TimeSeriesSplit
Five expanding-window folds; no future-to-past leakage.

03 Three classifiers
Logistic Regression, Random Forest and XGBoost.

04 Risk-aware scoring
Precision, recall, F1 and PR-AUC; FN cost = 5× FP.

SOURCES & STUDY BOUNDARIES

DATA SOURCES

Hong Kong EPD hourly AQHI (2021–2024); Hong Kong Observatory daily weather observations, merged by date.

INTERPRETATION

A strong recent-AQHI signal supports short-horizon warning, but does not establish causal effects of weather or station characteristics.

LIMITATIONS

Weather inputs are daily, not hourly; station locations do not align perfectly. AQHI methodology changed in 2025, so that period is excluded.

HELD-OUT TEST RESULTS

MODEL	THRESH.	PR-AUC	PRECISION	RECALL	F1
Logistic Regression	0.978	0.682	0.608	0.654	0.630
Random Forest	**0.893**	**0.683**	**0.542**	**0.738**	**0.625**
XGBoost	0.748	0.654	0.507	0.698	0.588

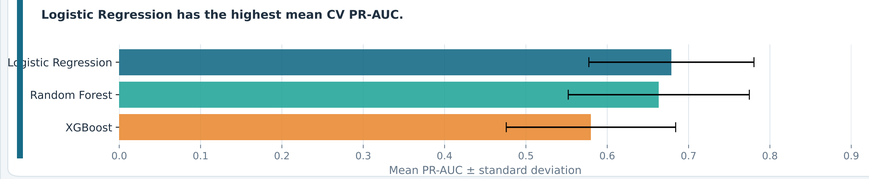
RF leads test recall and PR-AUC; LR gives the highest precision.

Thresholds selected on test probabilities (min recall 0.65; maximize F1).

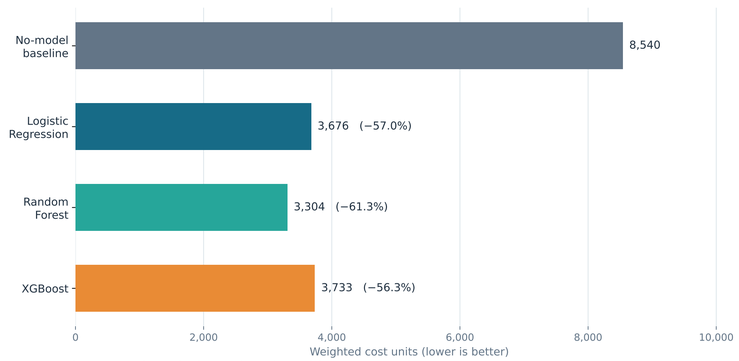
Treat threshold-dependent test metrics as exploratory; tune on validation data before deployment.

FIVE-FOLD TIME-SERIES CROSS-VALIDATION

Logistic Regression has the highest mean CV PR-AUC.



ALERT COST UNDER FN = 5 × FP

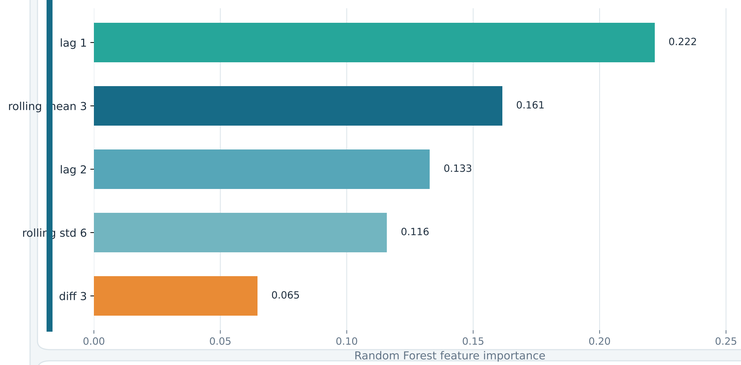


Random Forest: fewest missed alerts (448) and lowest total cost.

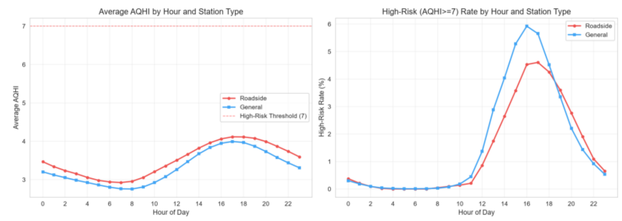
Cost assumption: each missed high-risk hour costs five false alarms.

WHAT DRIVES THE WARNING?

Recent AQHI levels dominate the model's feature importance.



WHEN IS HIGH RISK MOST COMMON?



PRACTICAL TAKEAWAY

Choose Random Forest when missed alerts matter most. Choose Logistic Regression when fewer false alarms and higher precision are the priority.

CAVEAT BEFORE DEPLOYMENT

The reported alert thresholds were selected using the held-out test probabilities. This makes threshold-based test scores exploratory, not unbiased final estimates.

Select thresholds on a validation period, then evaluate once on a separate untouched test period.